



• VST3 • AU • STANDALONE

MASTER BUS

COLOUR

THE DRUM RACK, UNCHAINED

The master rack of AudioXL DRUMS
as an effect plugin in its own
right.



USER MANUAL

1.0.0 VER

PRESETS

46





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ABOUT THE FIGURES

The figures are snapshots of the real window, retaken at every build: what you see here is what opens in your host. The enlarged controls are the Blender renders the plugin's skin is built from.

01 — GETTING STARTED

ANY TRACK, ANY BUS, THE MASTER

AudioXL Master Rack is the master rack of AudioXL DRUMS as a standalone effect plugin, in **VST3**, **Audio Unit** and **standalone application** form. Put it at the end of the chain – on a track, on a bus, on the master – and it brings the instrument's colouring to any material.

SAME STAGES

It is not a copy of the code: the stages are the same ones as in the instrument. What you hear here is what you hear inside DRUMS.



FIG. 01 The window: top bar, EQ, then the rack in two columns.

NOTE

On first launch the plugin runs as a demo: every thirty seconds it lets a breath of noise through. The serial goes in **PREFERENCES > LICENSE**.

02 — THE CHAIN

TOP TO BOTTOM, AS SOUND FLOWS

The order is that of the DRUMS rack, with four extra stages at the end. Read it from the top of the left column to the bottom of the right one: it is the order in which sound actually crosses the plugin.

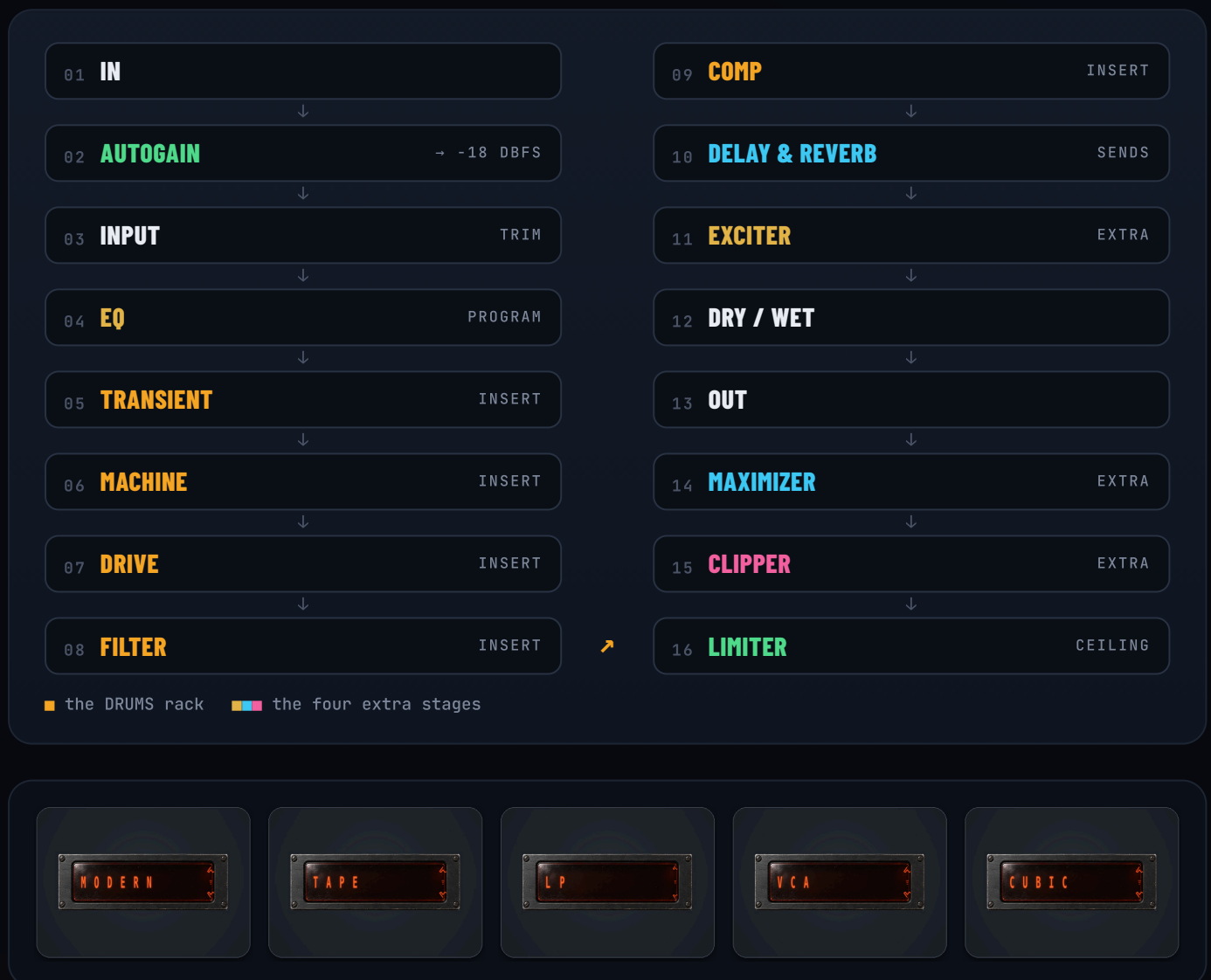


FIG. 02 The module displays, modelled and rendered in Blender: every stage states its own character on its little screen.

MODULE	WHAT IT DOES
EQ	A program equaliser in the Pultec EQP-1A style: three sections, stepped frequencies.
TRANSIENT	Independent attack and sustain, on differential envelopes. Bipolar : you can take away as well as add.
MACHINE	The 16 conversion models of the classic drum machines: bit depth, sample rate, bandwidth and saturation of each.
DRIVE	Eight characters , with a tilt EQ, parallel mix and noise keyed to the envelope.
FILTER	Multimode LP / HP / BP with resonance.
COMP	A linked stereo compressor with four characters , parallel mix and a reduction meter.
DELAY	A send synced to the host tempo, from parallel to ping-pong with WIDTH.
REVERB	A send with pre-delay and width. It taps before the chain: it is a send, not an insert.
EXCITER	It regenerates the top of the spectrum instead of lifting it with an EQ.
MAXIMIZER	A single control from 0 to 200 % : it saturates lows and highs and adds density, before the clipper.
CLIPPER	It takes the peaks off before the limiter, with five curves .
AUTOGAIN	At the top next to the input meter: it brings what comes in to -18 dBFS RMS , the level the thresholds are calibrated around.
LIMITER	A true ceiling with lookahead: the last thing before the output.
OUTPUT	INPUT, DRY/WET of the whole rack, OUTPUT.

03 — PROGRAM EQ

BOOST AND CUT THE SAME LOWS

Frequencies are chosen in steps, the rest is done by ear.

Three sections, as on the original unit, and no halfway house.



FIG. 03 The three stepped frequencies and the EQ knob, modelled on the front panel of the original unit.

SECTION	STEPS	WHAT IT DOES
LOWS	20 • 30 • 60 • 100 Hz	With a BOOST and an ATTEN on the same frequency. It looks like a contradiction and is in fact the reason that unit stayed in use: the two curves don't cancel because they don't have the same shape. Raise them together and the hump stays underneath while a dip opens in the low mids: bigger kick and less mud, in two knobs.
HIGHS	3 • 4 • 5 • 8 • 10 • 12 • 16 kHz	A bell with BANDWIDTH : narrow and tall is air, wide and low is presence.
HIGH ATTEN.	5 • 10 • 20 kHz	A separate shelf with its own frequency. Used together with the boost, it softens the top end without losing the sparkle.

UNDER THE HOOD

The curves are **RBJ biquads**, so what is written is what comes out. At the end a barely-there **valve stage**, blended rather than in series, proportional to how much you are boosting: it adds harmonics without compressing.

04 — DRIVE, COMP, EXCITER

EIGHT DRIVES, FOUR COMPRESSORS

DRIVE — EIGHT CHARACTERS

The first four are the ones from AudioXL DRUMS, in the same order and with the same code, so a preset arriving from there finds its character in its place.

TAPE

Tape: the most neutral, it compresses the peaks and that's all.

TUBE

Valve: asymmetric, even harmonics, it warms.

CLIP

Hard clip: aggressive, modern.

FOLD

Wavefolding: past a point the signal turns back on itself, it sounds electronic.

GERM

Germanium: like the old transistors, the negative half saturates first. It thickens.

DIODE

Diode: it compresses at once and never really clips. The gentlest of them.

RECT

Rectifier: half wave, which is an octave up mixed into the signal. Dirty.

CRUSH

Not a saturation at all, a broken converter: quantisation down to a few steps.

FROM
DRUMS

| TAPE**| TUBE****| CLIP****| FOLD**

NEW
HERE

| GERM**| DIODE****| RECT****| CRUSH**

COMP — FOUR CHARACTERS

The engine is the one from AudioXL DRUMS. The type doesn't change it, it changes **how it behaves** – which is after all the only thing that really tells two compressors apart. The ATTACK and RELEASE knobs stay the same, but each character rescales them: **8 ms on FET is not the 8 ms of VARI-MU**, exactly as between two real machines.

VCA

Solid state, identical to the DRUMS compressor. Precise: it does what the knobs say and nothing more.

OPTO

Optical: a wide knee and a release that slows down the harder it is compressing. You don't hear it working, you hear the track holding together.

FET

Attack four times faster, an almost hard knee, a more aggressive ratio and a thread of grit on the peaks.

VARI-MU

Valve: the ratio grows with level instead of being fixed, a very wide knee, slow times. It compresses a lot without you being able to tell where.

EXCITER

It extracts the band above **FREQ**, runs it through a harmonic generator and puts it back.

TUBE

Asymmetric, even harmonics: it warms and thickens.

TAPE

Symmetric and soft, odd harmonics: it tightens and polishes.

DIODE

A rectifier, the most aggressive of the three: air and glass on the hats.



05 — CLIPPER AND LIMITER

SHAVE THE PEAK, HOLD THE CEILING

The **clipper** sits before the limiter, not in its place, because they do two distinct jobs. The limiter chases the peak with a gain that rises and falls, and on a dense mix that movement is audible as breathing. The clipper has no gain: it takes the peak off and that is all, sample by sample, and **the two or three decibels the limiter no longer has to chase are the ones that hold the body of the track still.**

TRANSFER CURVES (ILLUSTRATIVE)

**HARD**

The peak disappears, odd harmonics, the hardest of them.

SOFT

Tanh: the peak rounds off much earlier, the most transparent.

CUBIC

Polynomial with a smooth join: the classic mix-bus middle ground.

FOLD

It folds instead of clipping: above threshold it becomes another sound.

DIODE

Asymmetric, clipping only below: even harmonics, a valve flavour.

The curves are normalised to unity slope around zero: **changing the type changes the colour, not the volume.** To push, use **DRIVE**, which changes how much is clipped and not how much comes out. **CEILING** is the ceiling of the stage, **MIX** sets how much of the clipped signal comes back in.

LIMITER

Lookahead fixed at **2 ms**, declared to the host as latency and compensated automatically. It is fixed on purpose: if it depended on a knob, latency would change while you play. The delay line runs with the limiter off too, so the compensation stays valid when you switch it on and off to compare.

CONTROL	RANGE	WHAT IT DOES
DRIVE	0 ... 18 dB	It pushes into the limiter: this is the perceived-loudness knob .
CEILING	dBFS	The true ceiling. A CEILING at -0.3 dB comes out at -0.3 dB: measured, not estimated .
RELEASE	10 ... 500 ms	How fast the gain comes back.

LOOKAHEAD

2 ms

fixed, compensated

DRIVE

0-18 dB

into the limiter

RELEASE

10-500

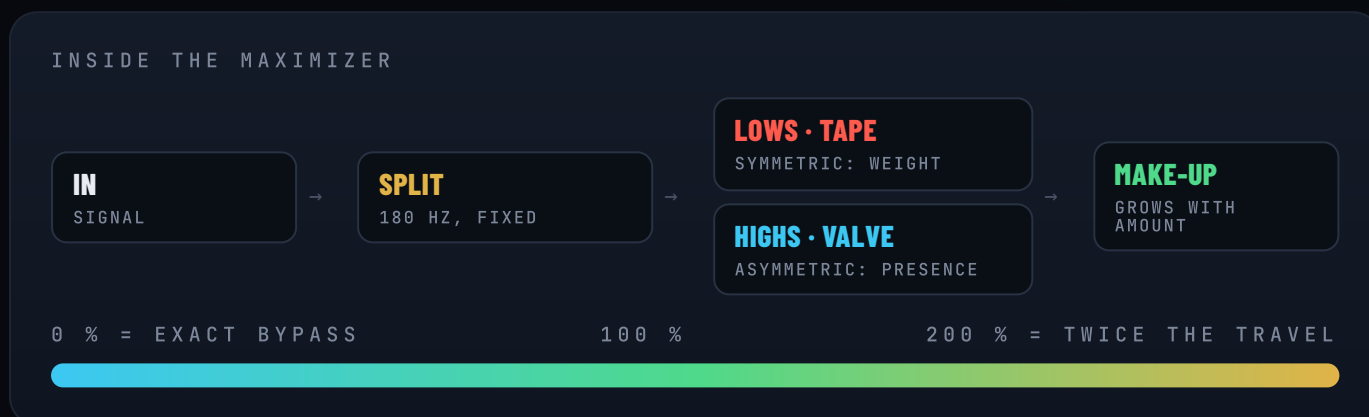
milliseconds



06 — MAXIMIZER

ONE KNOB, 0 TO 200 %

A single control, from 0 to 200 %. The idea is that of the master-bus “one-knob” plugins such as Cradle’s The God Particle: one knob that keeps pushing past 100 %, adding density and presence rather than simply saturating more. **It is not a clone** – inside that plugin there are an EQ, a multiband and a limiter which this rack already has as separate modules – only the control takes after that idea.



The recipe: the signal splits into two bands at a **fixed 180 Hz** crossover, not adjustable – this is one control, not another EQ to set up. The low band gets a symmetric **tape-like** saturation, which gives weight without darkening. The high band gets a slightly asymmetric **valve-like** saturation, which opens presence without turning metallic. A make-up gain that grows with AMOUNT makes the result fuller and not merely more saturated.



NOTE

At zero it is an exact bypass: switching the module off or leaving it at 0 % is the same thing. At 200 % it pushes twice the travel of 100 %, not four times – the scale is linear in AMOUNT, not in decibels. It sits before the clipper and is not a ceiling: it can push the signal past 0 dBFS, and keeping that in check is the job of the clipper and limiter that follow.

07 — LINKING WITH AUDIOXL DRUMS

TAKE THE CHAIN FROM DRUMS

The **SYNC DRUMS** button at the top links the plugin to the drum machine, if there is one in the session. Nothing needs configuring: AudioXL plugins talk to each other on their own, so **it works across separate processes too**. It is the same switch that DRUMS calls SYNC MASTER RACK: turn it on from either side and the two windows stay in agreement.



FIG. 04 The three switches on the top bar, off and on: RACK, SYNC DRUMS and the preset padlock.

SEPARATE

Each on its own: the drum machine keeps its rack.

WAITING

On, but there is no drum machine in the session yet. You can switch it on before opening DRUMS: when the link arrives, it engages by itself.

CHAIN TAKEN

Linked: from here on the rack doing the colouring is this one, and the DRUMS rack has stepped aside.

When it engages, the whole DRUMS master rack arrives here as it stands – transient, machine, drive, filter, compressor, delay and reverb, **values, menus and switches included** – and over there it goes into bypass. From that moment they are this plugin's knobs: move them as you like, nobody will put them back.

THE PADLOCK

Next to the preset menu there is a padlock, and it decides who is in charge.

OPEN

The rack follows AudioXL DRUMS: change preset or kit over there and it arrives here too. Knobs moved here stay where they are until something changes over there, so you can work without seeing them reset under your hands.

CLOSED

What you have built here is not replaced. The link hands the chain over as always – DRUMS switches its modules off, no double colouring – but nothing arrives on this side.

Open it again and the rack realigns at once: while it was closed the reference snapshot was not being updated, so the first thing it sees is a change. Like the sync and the autogain, **the padlock doesn't end up inside saved presets**: it is a choice about this session, not a sound.

WHAT TRAVELS AND WHAT DOESN'T

ARRIVES FROM DRUMS

- Transient
- Machine
- Drive
- Filter
- Compressor
- Delay
- Reverb

SWITCHED OFF ON LINK

- EQ
- Exciter
- Maximizer
- Clipper

switch them back on anytime

STAYS ON

- Limiter
- protection, not colour

EQ, exciter, maximizer and clipper **switch off** the moment the link engages: what has to start is the drum machine's chain, not that chain plus four stages left on by a preset chosen earlier. Switch them back on whenever you like. **The limiter stays**: it is protection, not colour.

NOTE

The **single-voice sends of DRUMS don't arrive** – the REVERB and DELAY knobs on the voice panel – because they have engines of their own, one per track, and they stay over there. The DRIVE and CLICK of each voice don't arrive either: they are controls inside that track's synthesis engine, not rack modules, and they stay per channel – the rack's DRIVE (on the whole bus) and the voice's DRIVE share a name but not a stage.

08 — AUTOGAIN

EVERY STAGE IN THE RIGHT PLACE

The compressor thresholds, the limiter knee and the amount of drive saturation are numbers in **absolute decibels**. A mix coming in at -30 dB doesn't even touch them; one coming in at -6 dB blows straight through. In both cases the same preset sounds different from what it was written for, and that is not the preset's fault.

The **AUTOGAIN** button, at the top next to the input meter, brings what comes in to **-18 dBFS RMS**, the level this chain's thresholds are calibrated around. From there on every stage works in the right place.



FIG. 05 The input meter in its four states: idle, working, hot, clipping.

Under the button you read how much it is correcting, with the sign, and towards which level. It sits next to the input meter on purpose: that is where you look at what is coming in, and that is where you need to know how far from working level the material was.

NOTE

It is not a compressor. The gain moves over a few seconds and doesn't chase the dynamics: it respects them. Below -60 dBFS it stops correcting, rather than lifting the noise floor by twenty decibels during a pause. It sits before everything, INPUT trim included, and doesn't touch the dry branch: with DRY/WET hard left the plugin still does nothing.

TWO DIRECTIONS, TWO SPEEDS

LOWERING · LOUD MATERIAL

Relatively quick

Too much level is dealt with fast.

RAISING · QUIET MATERIAL

Slow, in seconds

Deliberately unhurried.

That is what stops a simple transport stop from being read as *"the track is fading away, it needs more and more gain"*: press play again and the earlier material finds the earlier correction, not an extra push to work off.



09 — PRESETS AND PREFERENCES

46 PRESETS, SIX CATEGORIES

MASTER PRESET: the menu opens grouped – at these numbers a single list would be unreadable. Your own presets are saved into, and found again in, the same menu.



FIG. 06 The preset bar and the preferences button, off and on.

DRUM BUS

12

The effect profiles of AudioXL DRUMS, same names and same numbers: a preset with the same name sounds as it does inside the instrument.

BUS

4

To hold a subgroup together.

BUS GLUE PARALLEL LOUD BYPASS

ANALOG

8

The imprint of an analogue chain on any material – the colour, not the genre.

CONSOLE TAPE 15 IPS TAPE 7.5 IPS
TRANSFORMER VALVE CLASS A SUMMING
VINTAGE DESK

ELECTRONIC

8

Tuned for electronic genres, where the material arrives clean and digital and needs dirtying with some judgement.

BERLIN TECHNO DETROIT TECHNO DUB TECHNO
WARM HOUSE ACID DRUM & BASS MELODIC TRANCE
HARDGROOVE

SYNTHS

6

Instrument bus rather than mix bus.

SOLID BASS WIDE PAD LEAD UP FRONT ARP
STAB TEXTURE

MASTERING

8

For a whole mix: a light hand everywhere and a -1 dB ceiling, which is what the platforms expect.

WHAT A PRESET CARRIES

A preset is a **snapshot of the whole chain**, clipper and maximizer included. Three things stay out of it:

SYNC WITH DRUMS

A choice about this session.

THE PADLOCK

A choice about this session.

AUTOGAIN

A choice about what is coming in.

They are not sound, and a preset carrying them along would change the link or the level of whoever opens it without being asked.

PREFERENCES

The **OPTIONS** tab is where you choose the **window scale** and whether the **limiter should work with the rack switched off**. By default it does: the RACK switch is there to compare, not to remove the protection. If you want the raw comparison, switch it off here.



10 — AUTHORISATION

IT NEVER STOPS THE MUSIC

Without a serial, AudioXL Master Rack runs as a demo: **every thirty seconds it lets through four tenths of a second of noise**, with a bell in and out so that it doesn't click.

The plugin never stops: on a master bus, cutting the audio halfway through a take would be a cruel way to be remembered.

To unlock: **PREFERENCES > LICENSE**, paste the serial you received with your purchase and press **UNLOCK**. The noise disappears at once, with nothing to reopen.

NOISE EVERY

30 s

LASTING

0.4 s

AUDIO CUT

never

SERIAL FORMAT

AXR-XXXXX-XXXXX-XXXXX

Type it without worrying about case, dashes or spaces.

FIG. 07 The LICENSE tab in the preferences.

NOTE

The serial is registered on this computer: on another machine it has to be entered again. An AudioXL Master Rack serial will not open AudioXL DRUMS: they are two products with two different keys.



COLOUR ANYTHING

The master rack of AudioXL DRUMS,
on any track, any bus, the master.

